Sprint 4

Empirical Analysis Report:

From the Dataset to the Linker

Linker Adaptation to AIDev v5

October 2026

work in progress…

1 · AIDev

pull_request.parquet: 100% 57.6M/57.6M [00:03<00:00, 19.1MB/s]

3. First look

```
quick_look(prs)
```

[4] ✓ 0.4s

Rows: 71,677 Columns: 14

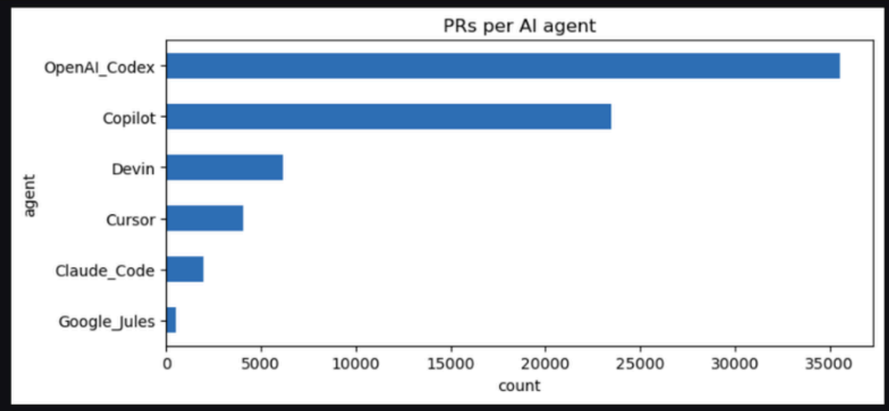
	id	number	title	body	agent	user_id	user	state	created_at	closed_at	merged_at
0	3369524944	9	feat: Add extensive list of EV charging tools and resources	This pull request resolves the issue by adding an extensive list of over 150...	Google_Jules	161369871	google-labs-jules[bot]	closed	2025-08-30T21:33:07Z	2025-08-30T21:40:30Z	2025-08-30T21:40:30Z
1	3369556773	10	Add last activity date to GitHub links	This change adds the date of the last activity to GitHub links in	Google_Jules	161369871	google-labs-jules[bot]	open	2025-08-30T22:20:29Z	None	None
2	320590845										2025-07-06T02:37:38Z

	type	missing_%	unique_values	example
id	int64	0.0	71677	[3369524944]
number	int64	0.0	21441	[9]
title	object	0.0	69452	[feat: Add extensive list of EV charging tools and resources]
body	object	0.8	70669	[This pull request resolves the issue by adding an extensive]
agent	object	0.0	6	[Google_Jules]
user_id	int64	0.0	3309	[161369871]
user	object	0.0	3309	[google-labs-jules[bot]]
state	object	0.0	2	[closed]
created_at	object	0.0	71203	[2025-08-30T21:33:07Z]
closed_at	object	8.3	65104	[2025-08-30T21:40:30Z]
merged_at	object	32.4	48365	[2025-08-30T21:40:30Z]
repo_id	int64	0.0	6673	[564884602]
repo_url	object	0.0	6673	[https://api.github.com/repos/juherr/awesome-ev-charging]
html_url	object	0.0	71677	[https://github.com/juherr/awesome-ev-charging/pull/9]

4. Which agents? How many PRs are merged?

```
if "agent" in prs.columns:
    plot_top(prs["agent"], title="PRs per AI agent")
if "state" in prs.columns:
    plot_top(prs["state"], title="PR state")
if "merged_at" in prs.columns:
    print(f"Merged: {prs['merged_at'].notna().mean():.1%}")
```

[5] ✓ 0.2s





AIDev versions

Data analysis – v3

```
C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5>uv run --project %EA% python step0_data_analysis_v3_v5.py
============================================================
AIDev v3 (AIDev-pop)
============================================================
PRs: 33,596 | repos: 2,807 | merged: 24,014 (71.5%)
created from 2024-12-24 to 2025-07-30
merged PRs in repos >=500 stars: 6,774
per agent (PRs, merged %):
   OpenAI_Codex       21,799    82.6%
   Copilot             4,970    43.0%
   Devin               4,827    53.8%
   Cursor              1,541    65.2%
   Claude_Code           459    59.0%
tables: ['VERSION.txt', 'human_pr_task_type.parquet', 'human_pull_request.parquet', 'pr_task_type.parquet', 'pull_request.parquet', 'repository.parquet']
```

6,774 merged PRs = the paper's number · Claude Code only 459 PRs

AIDev versions

Data analysis – v5

```
C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5>uv run --project %EA% python step0_data_analysis_v3_v5.py
============================================================
AIDev v5 (AIDev-7.6M)
============================================================
PRs: 361,296 | repos: 25,402 | merged: 246,777 (68.3%)
created from 2024-12-24 to 2026-03-31
merged PRs in repos >=500 stars: 141,710
per agent (PRs, merged %):
   Claude_Code       172,597    71.7%
   Copilot            98,384    60.6%
   OpenAI_Codex       62,731    75.2%
   Cursor             12,908    57.9%
   Devin              11,219    58.8%
   Google_Jules        3,457    60.0%
tables: ['VERSION.txt', 'pr_commit_details.parquet', 'pr_task_type_v5.parquet', 'pull_request.parquet', 'repository.parqu
merged PRs in repos >=500 stars, by language (top 5):
   TypeScript         44,195
   Python             22,496
   Go                 18,998
   Rust               12,513
   C#                  9,336
```

Claude Code 172,597 PRs · Python 22,496 merged PRs · no task type, no human PRs

Who Finishes the Job?

Takerngsaksiri et al. - 24 Sept 2026

22.9%
candidate fixes

4.5%
verified fixes

69.6%
fixed by the same agent

AIDev v3 · 30 days after merge

RQ1. How often are the merged agent PRs followed by a fix compared to the merged human PRs in the same repositories?
Result. 22.9 % of merged agent PRs are followed by a candidate fix and 4.5 % by a verified fix within 30 days. In the same repositories and the same observation window, agent merges are fixed at 1.41 times the human odds (candidate) and 1.62 times (verified), and the fixes arrive early: half of the 30-day incidence accumulates within the first week.

RQ2. Who authors the verified fix PRs of the merged agent PRs?
Result. 69.6 % of the verified fixes on agent merges come from the same agent, while 89 % of the verified fixes on human merges also come from humans. The finding shows that fixing work stays within the authoring population on both sides.

RQ3. To what extent are the commits pushed to the verified fix PRs authored by agents?

https://arxiv.org/pdf/2609.26847

The paper's study design

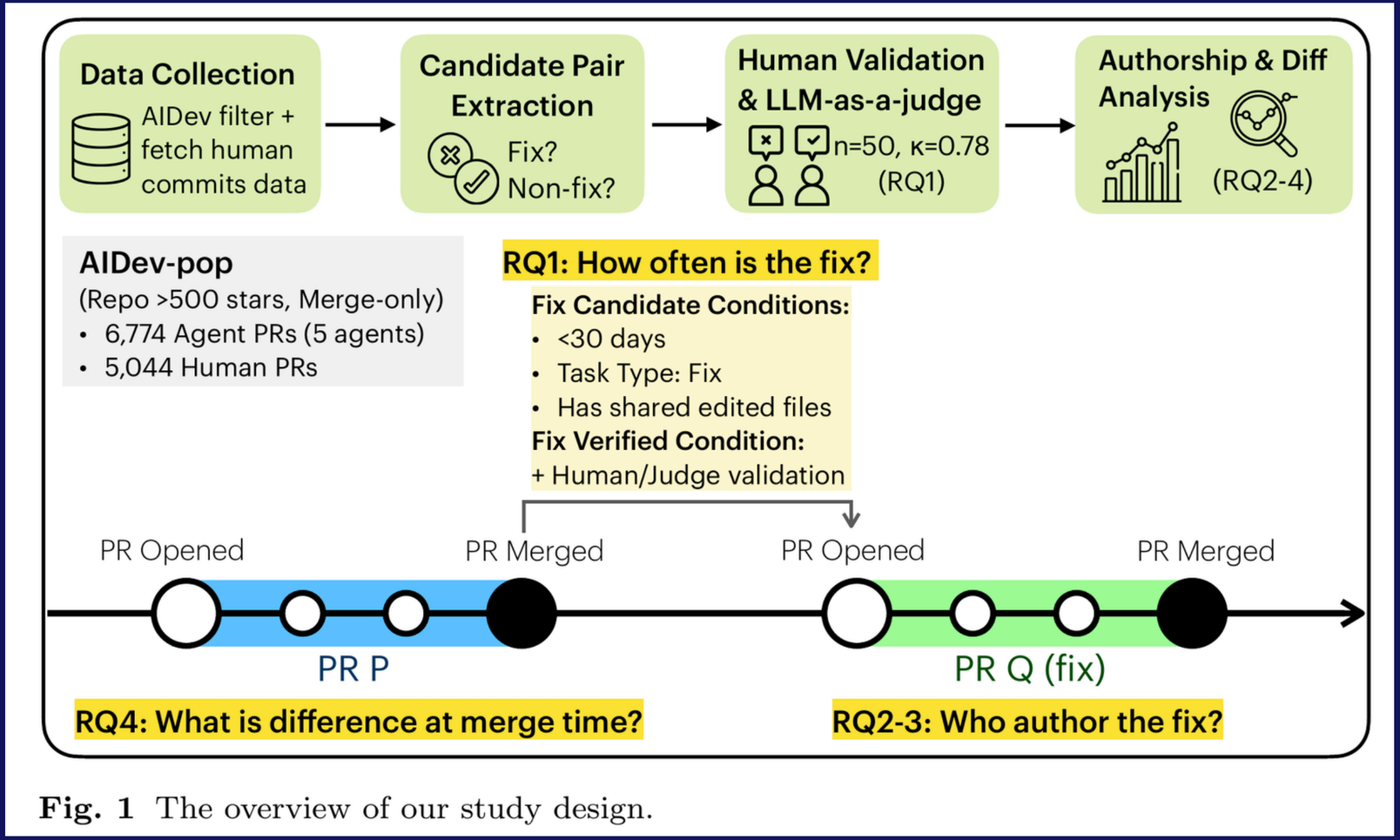


Fig. 1 The overview of our study design.

Takerngsaksiri et al. (2026), Fig. 1 – Section 3.1

https://arxiv.org/pdf/2609.26847

The tool that can help us: the linker

Input

Merged AI PRs (P) and all later PRs (Q) in the same projects

Linker

Simple rules on repo, dates, files and label – no code reading

Output

Pairs (P, Q): Q is a later change that may fix P

The linker is a script (linker.py) that, for each merged AI PR, automatically finds the later PRs in the same project that touch the same files soon after – the possible fixes of the AI code.

The original linker

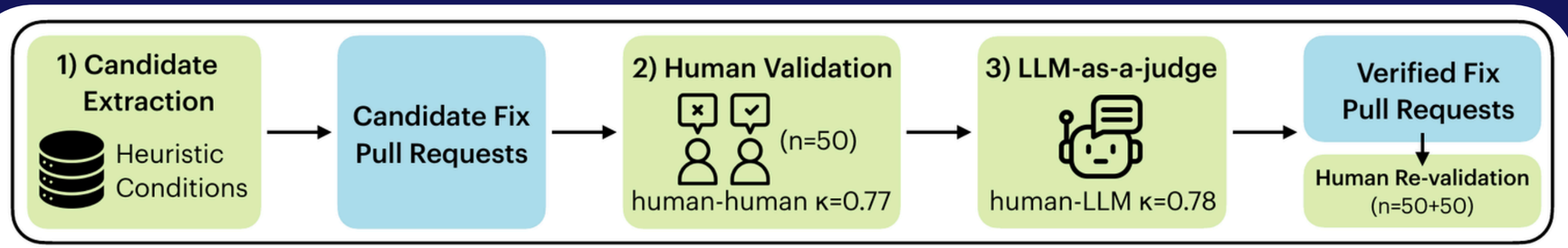


Fig. 2 (RQ1) The overview of the steps to identify verified fix PRs. A green box is a process; a blue box is a PR cohort.

1) Candidate Extraction. Given a merged PR P, our goal is to find a later *candidate* Q that fixes PR P. To do so, we apply four heuristic filters in order on both agent P and human P cohorts:

1. Q merges in the same repository, strictly after P, within 30 days.
2. Q is classified as `fix` type by the task-type tag in the AIDev dataset.
3. Q co-edits at least one file that P merged, identified by the intersection of net base-to-merge diff in both P and Q.
4. Q co-edited files include at least one non-boilerplate file. This is because files such as `lockfiles`, `changelogs`, and generated artifacts churn in nearly every PR.

Takerngsaksiri et al. (2026), Fig. 2 and Section 4.1.1

https://arxiv.org/pdf/2609.26847

The original linker

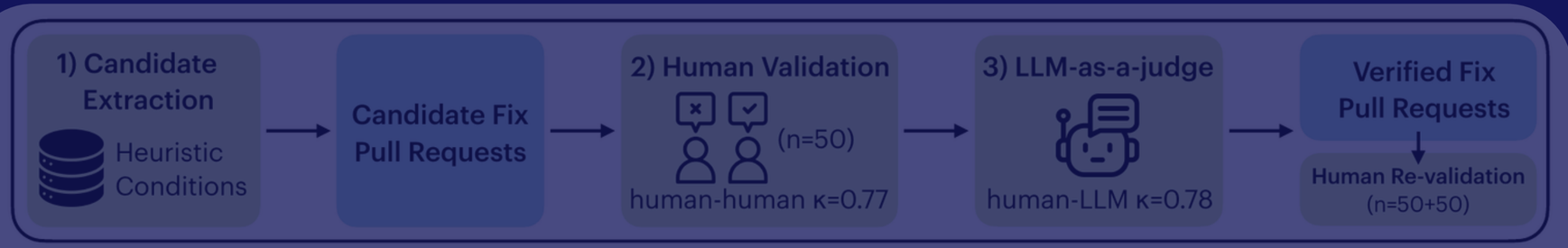


Fig. 2 (RQ1) The overview of the steps to identify verified fix PRs. A green box is a process;

Apply this Linker to the most recent data (AIDev V5)

candidate Q that fixes PR P. To do so, we apply four heuristic filters in order on both agent P and human P cohorts:

1. Q merges in the same repository, strictly after P, within 30 days.
2. Q is classified as `fix` type by the task-type tag in the AIDev dataset.
3. Q co-edits at least one file that P merged, identified by the intersection of net base-to-merge diff in both P and Q.
4. Q co-edited files include at least one non-boilerplate file. This is because files such as `lockfiles`, `changelogs`, and generated artifacts churn in nearly every PR.

Takerngsaksiri et al. (2026), Fig. 2 and Section 4.1.1

https://arxiv.org/pdf/2609.26847

Why not v3? Why not v5 as is?



v3

old data (to Jul 2025)
Claude Code: only 459 PRs

v5

no task-type table
no human PRs
→ linker cannot run

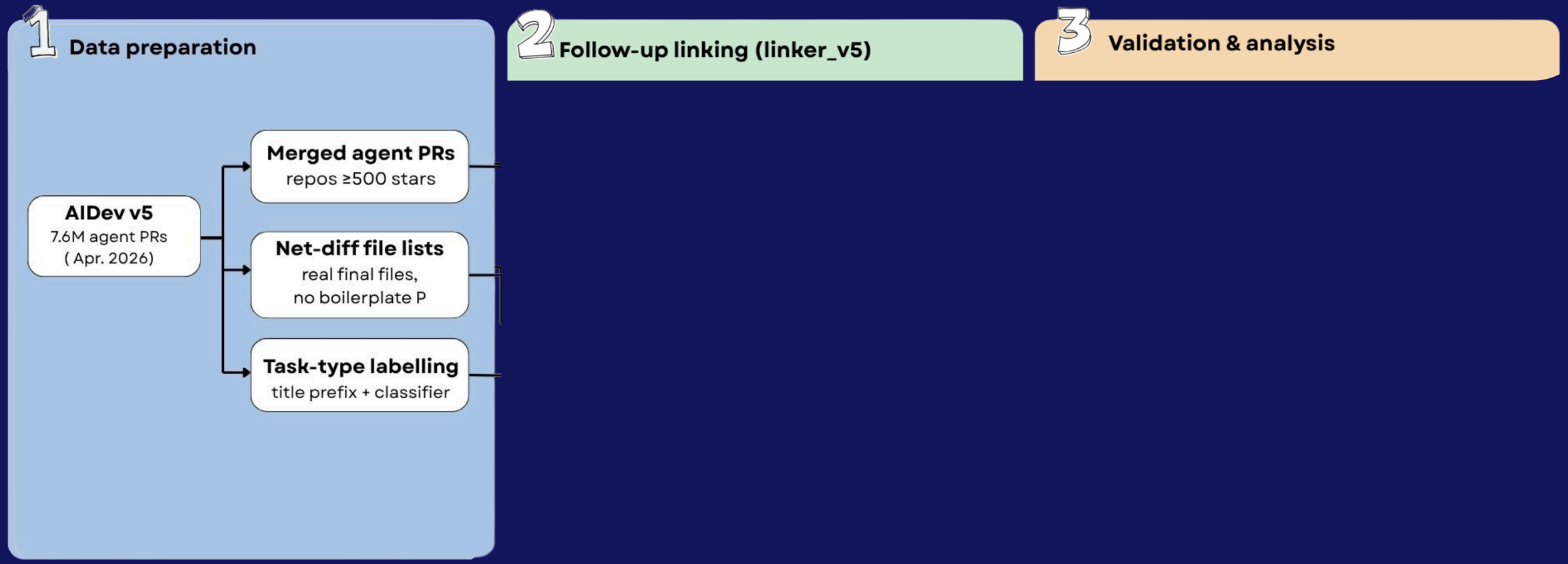
Linker_V5 Architecture

Stage 1: Data Preparation

1. Data preparation
2. Follow-up linking (linker_v5)
3. Validation & analysis

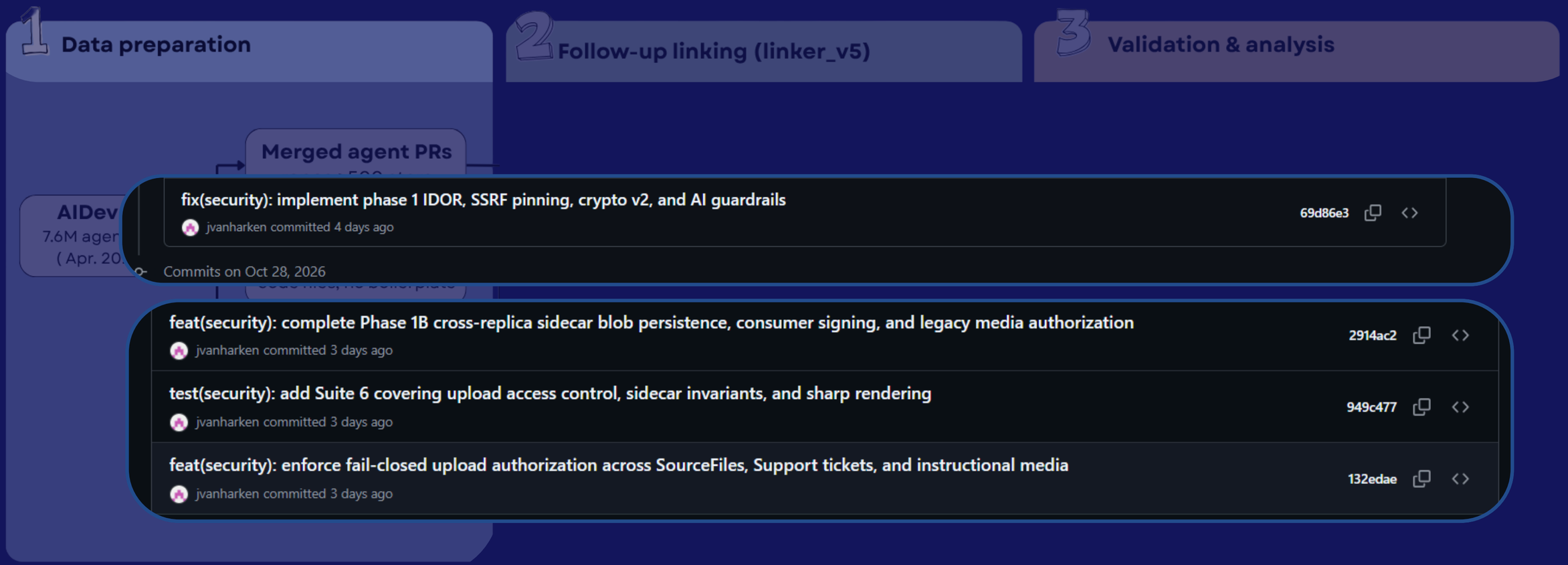
Linker_V5 Architecture

Stage 1: Data Preparation



Stage 1: read the title prefix

Stage 1: Data Preparation



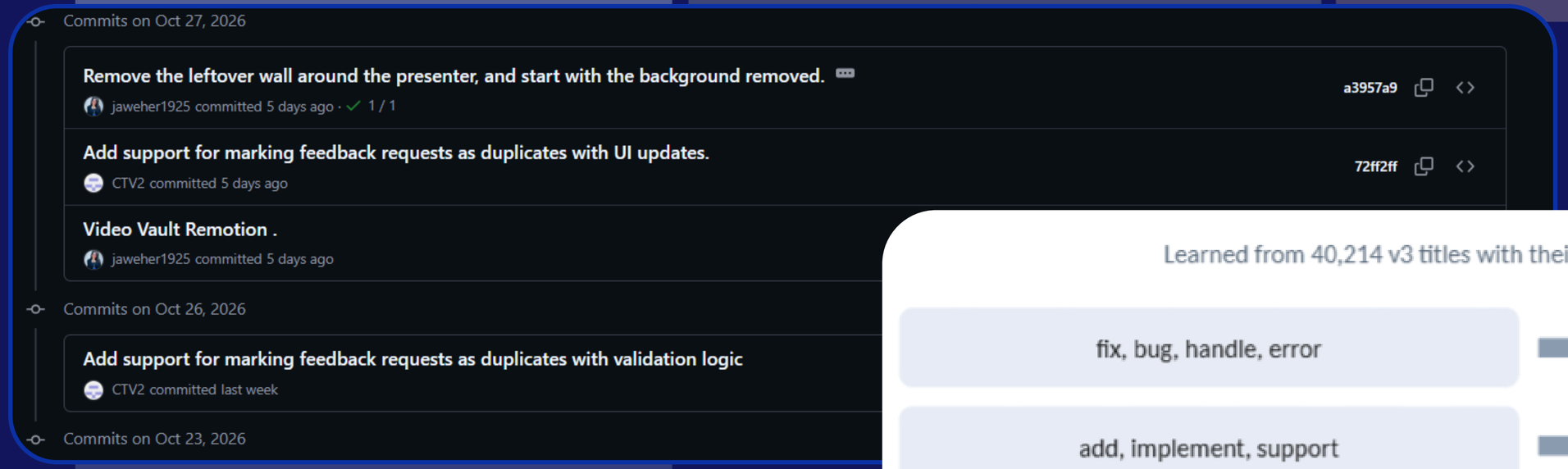
Step 2: A ML model guesses

Stage 1: Data Preparation

1 Data preparation

2 Follow-up linking (linker_v5)

3 Validation & analysis



Learned from 40,214 v3 titles with their real type

fix, bug, handle, error	fix
add, implement, support	feat
README, doc, typo	docs

The method is TF-IDF with logistic regression: it counts the important words, gives a score to each type, and the type with the highest score wins.

Result: our own table pr_task_type_v5

Stage 1: Data Preparation

```
Done -> C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\00_CODE\evolution-assurance\data\raw_v5
Next: step2_task_type_v5.py

=== STEP 2/6: rebuild the task type (title prefix + classifier) ===
v3 labels: 40,214 titles, types: {'feat': 16396, 'fix': 9884, 'docs': 4406, 'refactor': 2657, 'test': 2539, 'chore': 174
3, 'build': 1241, 'ci': 511, 'perf': 428, 'style': 247, 'other': 146, 'revert': 16}
  test on v3 - classifier only            : accuracy 84.2% | 'fix' precision 90.3%, recall 86.7%
  test on v3 - prefix + classifier (used) : accuracy 84.4% | 'fix' precision 90.8%, recall 86.4%

v5 PRs labelled: 361,296 (40% from the title prefix)
type
feat        146623
fix         122381
docs         28955
refactor     22765
test         12143
chore        11773
build         5669
ci            5228
perf          4160
style          791
revert         547
other          261

Saved C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\00_CODE\evolution-assurance\data\raw_v5\pr_task_type_v5.parquet
Next: step3_build_lake_v5.py
```

Stage 1: build the database

Stage 1: Data Preparation

1 Data preparation

2 Follow-up linking (linker_v5)

3 Validation & analysis

```
=== STEP 3/6: build the v5 database (merged agent PRs, code files) ===
Lake: C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\00_CODE\evolution-assurance\data\duckdb\aidev_v5.duckdb  (language: Pytho
n)
pr_files: 990,714 rows, 36,313 PRs, 2.9% boilerplate, 59.5% code files

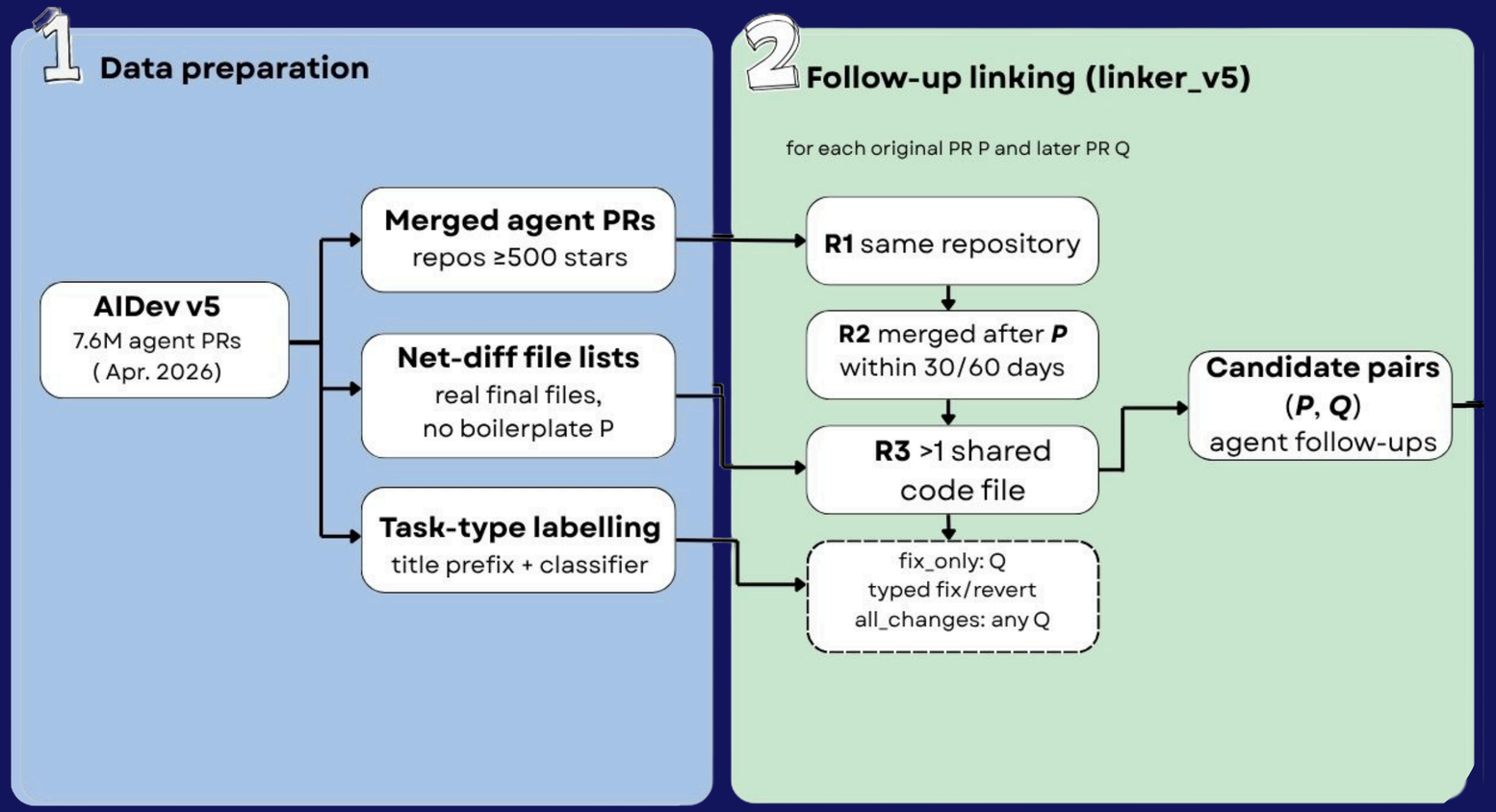
What linker_v5 will use (merged agent PRs, 500+ star repos):
       agent  merged_ai_prs  repos                first                latest
 Claude_Code          12534    942 2025-02-26T23:15:19Z 2026-04-06T12:54:37Z
     Copilot           3968    492 2025-05-20T21:07:01Z 2026-04-05T14:16:05Z
OpenAI_Codex           3491    364 2025-05-16T23:46:38Z 2026-04-06T18:19:41Z
       Devin           1622     28 2024-12-27T21:16:02Z 2026-04-02T20:29:10Z
      Cursor            827     70 2025-02-06T20:48:25Z 2026-04-02T23:30:28Z
Google_Jules             54     21 2025-07-13T12:29:44Z 2026-01-21T03:50:28Z

Next: step4_run_linker_v5.py
```

contains 22,496 merged Python agent pull requests in repositories with more than 500 stars.

Linker_V5 Architecture

Stage 2: Follow-up Linking



Two ways of counting

Stage 2: Follow-up Linking

all_changes

any later PR that passes the 3 rules
(new feature, refactor, fix …)

fix_only

only later PRs called "fix" or "Revert"
(title or our classifier)
= like the original

the linker does not read the code. It only looks at the project, the dates, the file names and the label.

Stage 2: run the linker

Stage 2: Follow-up Linking _ all_changes

C. === net diff, all_changes ===

RQ1 follow-up bug-fix rate per agent (N=22,496 merged PRs)

Agent	N PRs	% follow-up 30d	% follow-up 60d	% 30d (uncensored)	% 60d (uncensored)	mean # follow-ups 60d
Claude_Code	12,534	48.9%	51.7%	51.8%	56.1%	4.457
Copilot	3,968	33.6%	37.4%	34.7%	39.6%	2.053
OpenAI_Codex	3,491	53.0%	55.2%	55.4%	57.6%	10.067
Devin	1,622	38.1%	42.4%	40.7%	44.7%	2.189
Cursor	827	27.9%	32.4%	28.1%	34.7%	2.079
Google_Jules	54	24.1%	24.1%	24.1%	24.1%	0.370

```
[06:00:50] wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_all_changes_30_60
           \per_pr.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_all_changes_30_60
           \per_agent.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_all_changes_30_60
           \candidate_pairs.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_all_changes_30_60
           \candidate_pairs.parquet
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_all_changes_30_60
           \run_metadata.json
7,078 of 21,776 PRs in the time/repo candidate pairs have no file rows (net_sub); their pairs cannot be file-matched (no
rows in v5 pr_commit_details).
Candidate follow-up changes are identified through temporal and file-level relationships. These relationships do not
establish corrective intent or demonstrate that the original AI-generated contribution introduced a deficiency. Such
claims require subsequent code-level analysis and validation.
```

Any later AI change to the same code counts as a follow-up.

Stage 2: run the linker

Stage 2: Follow-up Linking _ Fix_only

C. === net diff, fix_only ===

RQ1 follow-up bug-fix rate per agent (N=22,496 merged PRs)

Agent	N PRs	% follow-up 30d	% follow-up 60d	% 30d (uncensored)	% 60d (uncensored)	mean # follow-ups 60d
Claude_Code	12,534	32.3%	35.7%	34.9%	40.0%	1.439
Copilot	3,968	19.9%	22.9%	20.3%	24.0%	0.641
OpenAI_Codex	3,491	34.5%	36.3%	38.4%	41.3%	3.392
Devin	1,622	19.1%	25.5%	20.5%	26.2%	0.485
Cursor	827	12.0%	15.8%	12.4%	17.8%	0.544
Google_Jules	54	11.1%	11.1%	11.1%	11.1%	0.130

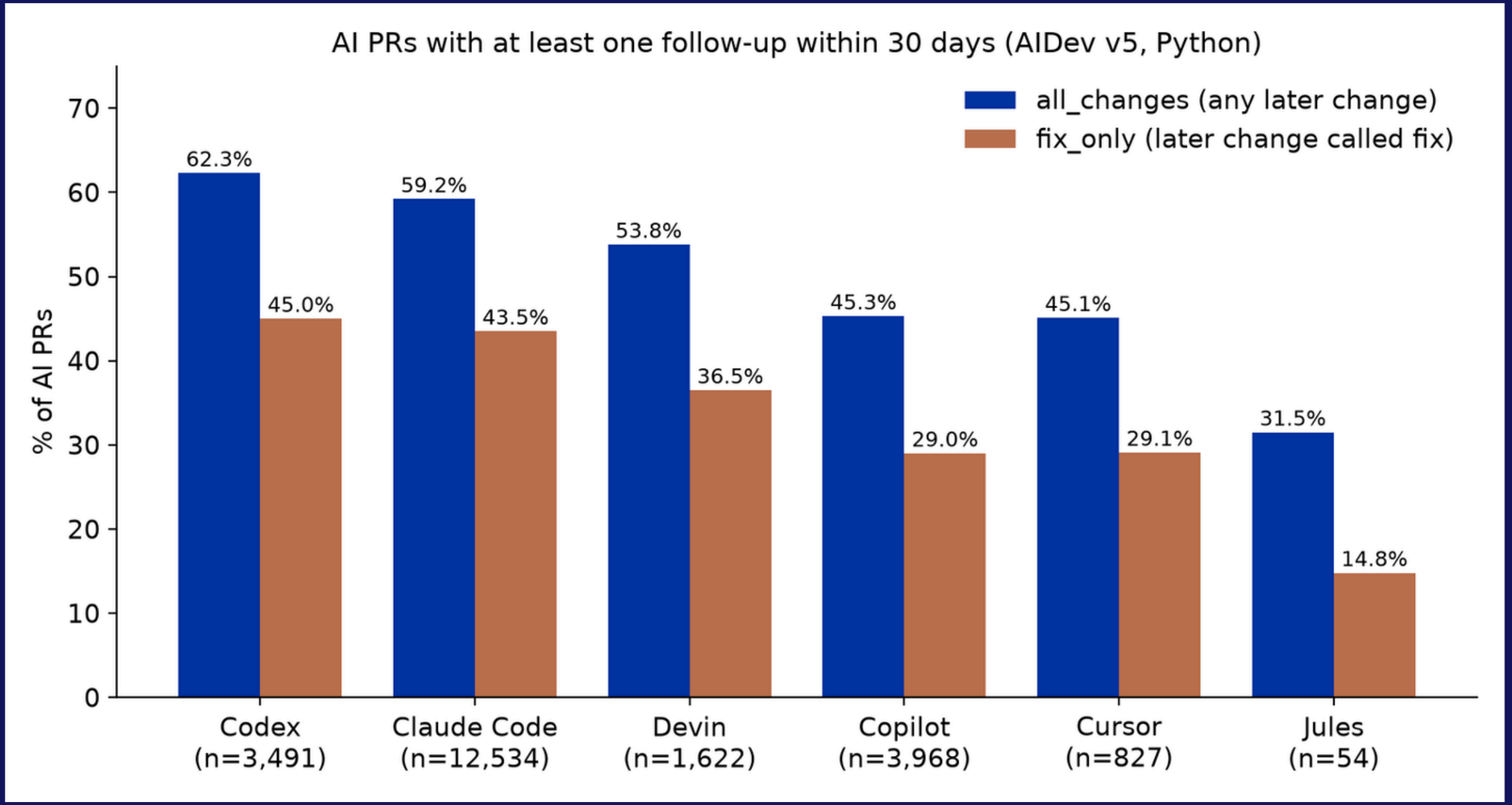
```
[06:00:54] wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_fix_only_30_60\pe
           r_pr.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_fix_only_30_60\pe
           r_agent.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_fix_only_30_60\ca
           ndidate_pairs.csv
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_fix_only_30_60\ca
           ndidate_pairs.parquet
           wrote                                                    linker_v5.py:821
           C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\results\net_fix_only_30_60\ru
           n_metadata.json
5,888 of 19,475 PRs in the time/repo candidate pairs have no file rows (net_sub); their pairs cannot be file-matched (no
rows in v5 pr_commit_details).
Candidate follow-up changes are identified through temporal and file-level relationships. These relationships do not
establish corrective intent or demonstrate that the original AI-generated contribution introduced a deficiency. Such
claims require subsequent code-level analysis and validation.
```

Only later changes labelled as a fix count.

Result of the linker

Result 1: follow-up rate per agent

① % of AI PRs per agent with a follow-up within 30 days



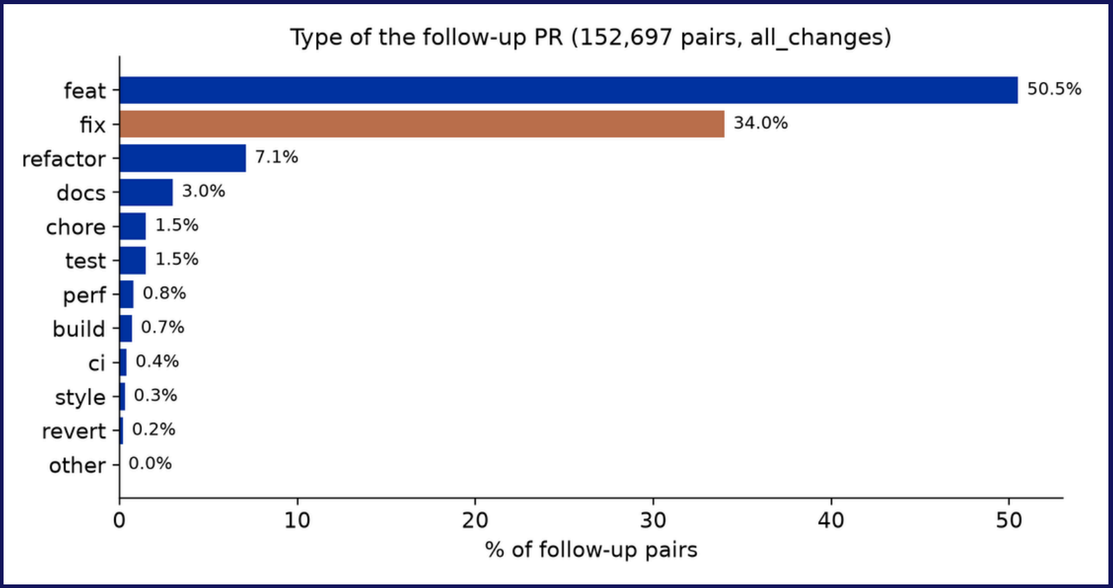
Result of the linker
Result 2: the list of pairs

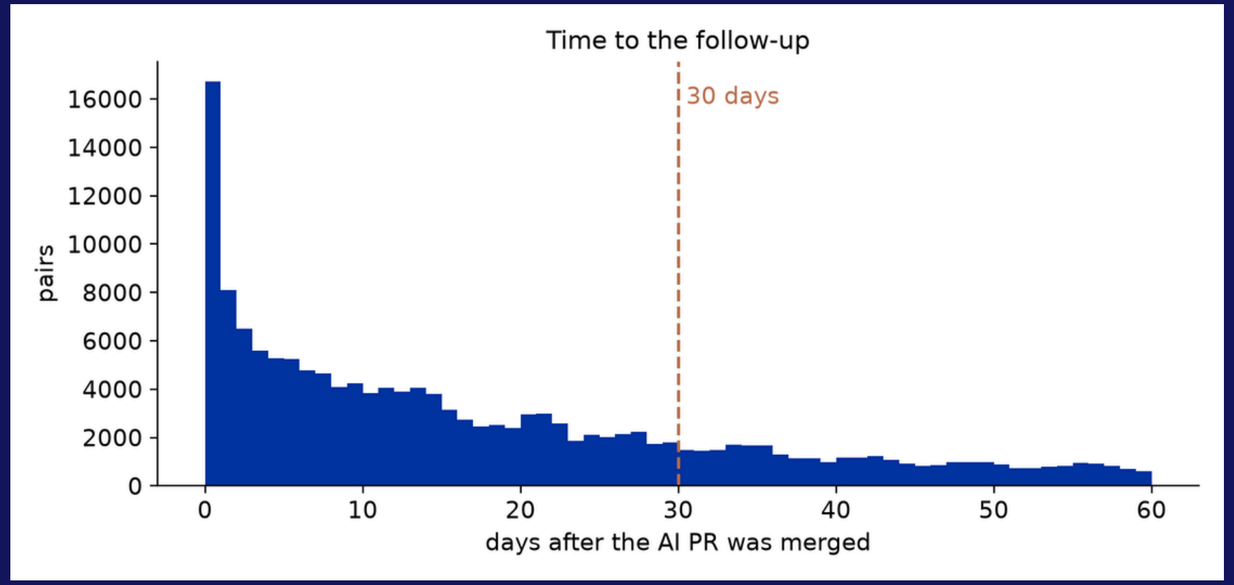
② the list of pairs: AI PR → follow-up PR, with the shared files

	original_agent	ai_pr_url	ai_pr_title	followup_agent	followup_url	followup_title	followup_task_type	days_after_merge
87690	Claude_Code	https://github.com/incidentfox/incidentfox/pull/19	feat(agent): Port ask_human and context-passing improvem...	Claude_Code	https://github.com/incidentfox/incidentfox/pull/429	feat(ai_pipeline): LLM-powered knowledge extraction for ...	feat	27.268762
103518	Copilot	https://github.com/ace-step/ACE-Step-1.5/pull/221	Fix 4B model support for 16GB GPUs by increasing VRAM de...	Claude_Code	https://github.com/ace-step/ACE-Step-1.5/pull/210	feat: Add AMD ROCm support for Windows (RX 7000/6000 ser...	feat	0.851204
20535	OpenAI_Codex	https://github.com/mfat/sshpilot/pull/217	Ensure quit dialog closes only once before releasing hold	OpenAI_Codex	https://github.com/mfat/sshpilot/pull/240	Detect macOS third-party terminals to show external options	feat	1.186944
130575	OpenAI_Codex	https://github.com/rmusser01/tldw_server/pull/791	Codex/docs only speech hygiene clean	OpenAI_Codex	https://github.com/rmusser01/tldw_server/pull/871	fix(ui): reuse redesigned speech playground for /tts	fix	10.816574

result is the list of pairs: each AI pull request with its follow-up and the shared files.

What the follow-ups look like

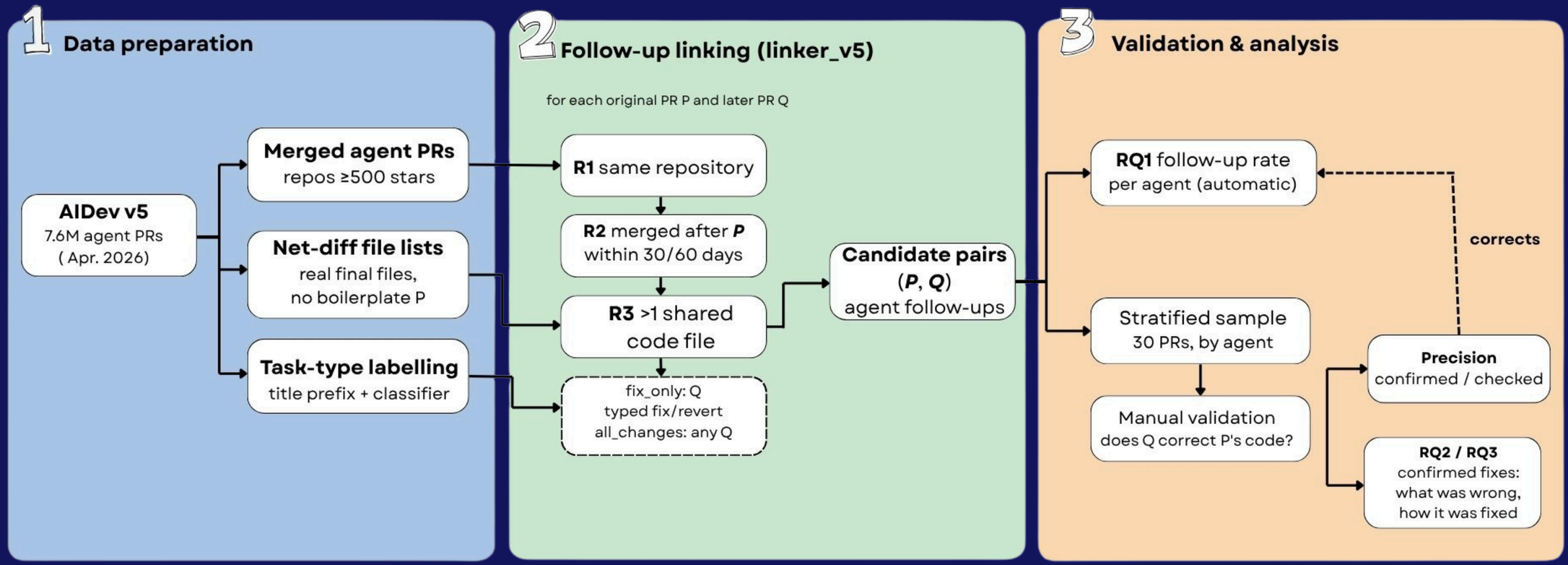




Most follow-ups come in the first month

Linker_V5 Architecture

Stage 3: sample for the manual check



Linker_V5 Architecture

Stage 3: sample for the manual check

```
C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5>uv run --python 3.13 --project %EA% --with openpyxl pytho
n step6_test_30_prs.py
pairs from results/net_all_changes_30_60
30 AI PRs, 68 pairs -> C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\Test_30_PRs_netdiff.xlsx
original_agent
Claude_Code     15
OpenAI_Codex     5
Copilot          4
Cursor           2
Devin            2
Google_Jules     2

C:\Users\GIGABYTE\Des
```

C2 | https://github.com/saulpw/visidata/pull/3005

test_id	ai_agent	ai_pr_url	ai_pr_title	followup_url	followup_title	followup_agent	followup_type	days_after	shared_files	1_same_code_changed? (yes/no)	2_corrects_the_AI_code? (yes/no)	3_what_was_wr
T01.1	Claude_Code	https://github.com/saulpw/visidata/pull/3005	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3007	Clean up test output and fix startpos	Claude_Code	fix	4,7	dev/STYLE.md; dev/TESTING.md; visidata/features/replay_bulk.py			
T01.2	Claude_Code	https://github.com/saulpw/visidata/pull/3005	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3019	Add top-level Makefile; stop checking in generated man pages	Claude_Code	feat	8,6	dev/TESTING.md			
T01.3	Claude_Code	https://github.com/saulpw/visidata/pull/3005	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3016	Add Column.getFullDisplayValue() for untruncated display values	Claude_Code	feat	11,9	dev/STYLE.md; visidata/loaders/fixed_width.py			
T02.1	Claude_Code	https://github.com/PrefectHQ/prefect/pull/20158	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20167	Add Template & Renderer for Custom Deployments SDK	Claude_Code	feat	3,7	plans/2026-01-06-custom-deployment-sdks.md; src/prefect/_sdk/naming.py; tests/_sdk/test_naming.py			
T02.2	Claude_Code	https://github.com/PrefectHQ/prefect/pull/20158	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20172	Add Data Fetching & Generator for Custom Deployments SDK	Claude_Code	feat	3,9	plans/2026-01-06-custom-deployment-sdks.md			
T02.3	Claude_Code	https://github.com/PrefectHQ/prefect/pull/20158	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20173	Add CLI Command for Custom Deployments SDK	Claude_Code	feat	4,8	plans/2026-01-06-custom-deployment-sdks.md			
T03.1	Claude_Code	https://github.com/lovit/soynlp/pull/222	docs(word): WordExtractor.R_suffix 역할 문서화	https://github.com/lovit/soynlp/pull/300	fix(word): get_entropy 전부 0인 리스트 입력 시 ZeroDivisionError 방어 가드 추가	Claude_Code	fix	1	soynlp/word/word.py			
T04.1	Claude_Code	https://github.com/pipecat-ai/pipecat/pull/3893	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/3967	Fix Azure STT SpeechConfig with private endpoint	Claude_Code	fix	7,3	src/pipecat/services/azure/stt.py			
T04.2	Claude_Code	https://github.com/pipecat-ai/pipecat/pull/3893	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/3974	Make Azure STT region optional when private_endpoint is used	Claude_Code	feat	7,9	src/pipecat/services/azure/stt.py			
T04.3	Claude_Code	https://github.com/pipecat-ai/pipecat/pull/3893	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/4024	Fix Language enum conversion at init time in base TTS/STT services	Claude_Code	fix	14,3	src/pipecat/services/azure/stt.py; src/pipecat/services/azure/tts.py			
T05.1	Claude_Code	https://github.com/marin-community/marin/pull/3161	Fall back to local compilation cache when MARIN_PREFIX is S3	https://github.com/marin-community/marin/pull/3177	Use GCS compilation cache for CW canary ferry	Claude_Code	feat	0,3	lib/marin/src/marin/training/training.py			
T05.2	Claude_Code	https://github.com/marin-community/marin/pull/3161	Fall back to local compilation cache when MARIN_PREFIX is S3	https://github.com/marin-community/marin/pull/3195	Enable S3 compilation cache and disable XLA autotune sub-cache	Claude_Code	feat	1,1	lib/marin/src/marin/training/training.py			
T05.3	Claude_Code	https://github.com/marin-community/marin/pull/3161	Fall back to local compilation cache when MARIN_PREFIX is S3	https://github.com/marin-community/marin/pull/3217	Fix CW canary OOM and improve training observability	Claude_Code	fix	1,4	lib/marin/src/marin/training/training.py			
T06.1	Claude_Code	https://github.com/basicmachines-co/basic-memory/pull/473	fix: detect rclone version for --create-empty-src-dirs support	https://github.com/basicmachines-co/basic-memory/pull/502	refactor: composition roots, deps split, and typed API clients (#490 roadmap)	Claude_Code	refactor	11	test-int/mcp/test_lifespan_shutdown_sync_task_c			

Manual check: result

A15 | T06.1

	ai_pr_title	followup_url	followup_title	followup_agent	followup_type	days_after	shared_files	1_same_code_changed? (yes/no)	2_corrects_the_AI_code? (yes/no)	3_what_was_wrong (Q2)	4_how_it_was_fixed (Q3)	notes
'pull	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3007	Clean up test output and fix startpos	Claude_Code	fix	4,7	dev/STYLE.md; dev/TESTING.md; visidata/features/replay_bulk.py	no	no			Same file, other code (Q does not touch P's lines)
'pull	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3019	Add top-level Makefile; stop checking in generated man pages	Claude_Code	feat	8,6	dev/TESTING.md	no	no			Same file, other code (Q does not touch P's lines)
'pull	Fix nested freq table dive-selected corrupting source chain	https://github.com/saulpw/visidata/pull/3016	Add Column.getFullDisplayValue() for untruncated display values	Claude_Code	feat	11,9	dev/STYLE.md; visidata/loaders/fixed_width.py	no	no			Same file, other code (Q does not touch P's lines)
t/pu	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20167	Add Template & Renderer for Custom Deployments SDK	Claude_Code	feat	3,7	plans/2026-01-06-custom-deployment-sdks.md; src/prefect/_sdk/naming.py; tests/_sdk/test_naming.py	yes	YES	P reserved the wrong names (run, run_async, with_options, with_infra) and missed 'self' → generated SDK code could break method signatures	Q replaced the reserved list with {'self'} and updated the tests	REAL FIX (pre-check)
t/pu	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20172	Add Data Fetching & Generator for Custom Deployments SDK	Claude_Code	feat	3,9	plans/2026-01-06-custom-deployment-sdks.md	no	no			Same file, other code (Q does not touch P's lines)
t/pu	Add Naming & Data Models for Custom Deployment SDK	https://github.com/PrefectHQ/prefect/pull/20173	Add CLI Command for Custom Deployments SDK	Claude_Code	feat	4,8	plans/2026-01-06-custom-deployment-sdks.md	no	no			Same file, other code (Q does not touch P's lines)
/222	docs(word): WordExtractor.R_suffix 역할 문서화	https://github.com/lovit/soynlp/pull/300	fix(word): get_entropy 전부 0인 리스트 입력 시 ZeroDivisionError 방어 가드 추가	Claude_Code	fix	1	soynlp/word/word.py	no	no			Same file, other code (Q does not touch P's lines)
	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/3967	Fix Azure STT SpeechConfig with private endpoint	Claude_Code	fix	7,3	src/pipecat/services/azure/stt.py	no	no			Same file, other code (Q does not touch P's lines)
	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/3974	Make Azure STT region optional when private_endpoint is used	Claude_Code	feat	7,9	src/pipecat/services/azure/stt.py	no	no			Same file, other code (Q does not touch P's lines)
	Propagate Azure TTS/STT cancellation errors to the pipeline	https://github.com/pipecat-ai/pipecat/pull/4024	Fix Language enum conversion at init time in base TTS/STT services	Claude_Code	fix	14,3	src/pipecat/services/azure/stt.py; src/pipecat/services/azure/tts.py	no	no			Same file, other code (Q does not touch P's lines)
	Fall back to local compilation cache when MARIN_PREFIX is S3	https://github.com/marin-community/marin/pull/3177	Use GCS compilation cache for CW canary ferry	Claude_Code	feat	0,3	lib/marin/src/marin/training/training.py	no	no			Same file, other code (Q does not touch P's lines)

run_v5\Test_30_PRs_netdiff_checked.xlsx

NB : I'd like to check them with you and agree on those result

```
[NbConvertApp] Converting notebook step5_results_v5.ipynb to html
C:\Users\GIGABYTE\AppData\Local\uv\cache\archive-v0\2qOmMs0gkcANV0UX\Lib\site-packages\zmq\_future.py:718: RuntimeWarnin
g: Proactor event loop does not implement add_reader family of methods required for zmq. Registering an additional selec
tor thread for add_reader support via tornado. Use `asyncio.set_event_loop_policy(WindowsSelectorEventLoopPolicy())` to
avoid this warning.
  self._get_loop()
[IPKernelApp] WARNING | Kernel is running over TCP without encryption. All communication (including code and outputs) is
 sent in plain text and is susceptible to eavesdropping. Use IPC transport or launch with kernel manager-provisioned Cur
veZMQ keys to enable transport encryption.
[NbConvertApp] WARNING | Alternative text is missing on 3 image(s).
[NbConvertApp] Writing 450111 bytes to step5_results_v5.html

=== STEP 6/6: make the 30-PR test sheet (Test_30_PRs.xlsx) ===
Test_30_PRs.xlsx already exists -> keeping it, writing Test_30_PRs_NEW.xlsx
30 AI PRs, 68 pairs -> C:\Users\GIGABYTE\Desktop\THESIS_Jaweher\Sprint4_Linker\run_v5\Test_30_PRs_NEW.xlsx
original_agent
Claude_Code     15
OpenAI_Codex     5
Copilot          4
Cursor           2
Devin            2
Google_Jules     2

======================================================================
 DONE. Open:
   step5_results_v5.html  - the result tables and charts
   Test_30_PRs.xlsx       - the 30 PRs to check by hand
======================================================================
Appuyez sur une touche pour continuer...
```